

# Exploratory Data Analysis on Retail Sales

## **1. Objective**

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Superstore retail sales dataset to identify sales trends, customer purchasing behavior, regional performance, and business insights using Python.

## **2. Tools and Technologies**

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

## **3. Dataset**

Dataset: Sample Superstore Dataset

The dataset contains retail sales transactions, including customer details, product information, sales, profit, discount, quantity, category, sub-category, and regional information.

## **4. Data Preprocessing**

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Checked dataset dimensions.
- Examined data types.
- Checked for missing values.
- Removed duplicate records.
- Converted the Order Date column into datetime format.
- Generated descriptive statistics for numerical columns.

## **5. Exploratory Data Analysis**

The following analyses were performed:

- Sales by Category
- Sales by Region
- Profit by Category
- Top 10 Sub-Categories by Sales
- Monthly Sales Trend
- Correlation Heatmap
- Customer Segment Analysis
- Sales Distribution
- Profit Distribution

## **6. Key Findings**

- Technology generated the highest sales revenue.
- Technology also recorded the highest overall profit.
- The West region achieved the highest sales among all regions.
- The South region recorded the lowest sales.
- Phones were the best-selling sub-category.
- Furniture generated comparatively lower profit despite strong sales.
- Higher discounts were associated with lower profits.

## **7. Business Insights**

- Focus marketing campaigns on Technology products to maximize revenue.
- Improve sales performance in the South region through targeted promotions.
- Review Furniture pricing and discount strategies to improve profitability.
- Optimize discount policies since excessive discounts reduce profit.
- Maintain sufficient inventory for high-demand products such as Phones.

## **8. Conclusion**

This project successfully analyzed the retail sales dataset using Python and data visualization libraries. The analysis identified significant sales patterns, profitable product categories, and regional performance differences. The insights obtained from

this project can support better business decision-making, inventory planning, pricing strategies, and marketing initiatives.

## **9. Future Scope**

- Build a sales forecasting model using Machine Learning.
- Develop an interactive dashboard using Power BI or Tableau.
- Predict customer purchasing behavior.
- Perform customer segmentation for personalized marketing.

## **10. Output**

The project generated:

- Cleaned dataset
- Statistical summary
- Sales visualizations
- Profit analysis
- Correlation heatmap
- Business insights and recommendations